
Bahadır Utku Kesgin

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Education

Koç University	Istanbul, Turkey
B.Sc., Electrical and Electronics Engineering	CGPA: 3.46/4.00
Track: Electronics and Waves	Track GPA: 4.00/4.00
Track: Quantum and Photonics	Track GPA: 3.85/4.00
Comprehensive National Athlete Scholarship (Turkish Ministry of Youth and Sports)	Expected Graduation: June 2027

Research Interests

Nonlinear and ultrafast photonics; optical and neuromorphic computing; photonic neural networks and reservoir computing; spatiotemporal dynamics in multimode fibers for scalable optical information processing.

Research and Professional Experience

Koç University — Laboratory of Photonic Technologies (PI: Dr. Uğur Teğin)	Istanbul, Turkey
<i>Undergraduate Researcher</i>	Feb 2023 – Present
- Lead author on optical-computing architectures that exploit nonlinear and chaotic spatiotemporal dynamics in multimode fibers (3 published first-author journal articles; 9 conference presentations). - Developed and characterized ultrafast single- and multi-mode fiber laser sources, and single-shot ultrafast imaging via spatial multiplexing (685 Gfps). - Built photonic neural-network and reservoir-computing frameworks (spiking, recurrent, and diffractive) in Python/PyTorch, validated both numerically and experimentally.	
PROFEN — Vacuum Electron Device Technologies Department	Istanbul, Turkey
<i>R&D Intern</i>	July 2025 – Sept 2025
Researched photocathode-based High-Power Microwave (HPM) sources and RF/THz ranging and detection technologies.	
Aquila Games	Remote
<i>Game and AI Programmer</i>	March 2020 – Oct 2022
Engineered NPC AI and core gameplay systems in C# / Unity Engine for <i>The Sentinel Remake</i> and <i>Neon Parkour</i>.	

Journal Articles

- Kesgin, B.U.** & Teğin, U. (2025). Photonic neural networks at the edge of spatiotemporal chaos in multimode fibers. *Nanophotonics* 14(16), 2723–2732. [\[DOI\]](#) ([Special Issue: Guided Nonlinear Optics for Information Processing](#)).
- Kesgin, B.U.** & Teğin, U. (2024). Implementing the analogous neural network using chaotic strange attractors. *Communications Engineering* 3, 99. [\[DOI\]](#) ([Nature Portfolio "Neuromorphic Hardware and Computing" Collection](#)).
- Kesgin, B.U.**, Yüce, F. & Teğin, U. (2025). Fiber-based diffractive deep neural network. *Optics Letters*. [\[DOI\]](#)
- Kesgin, B.U.**, Durdu, G.Y. & Teğin, U. (2025). Optical spiking neural networks via rogue-wave statistics. *npj Unconventional Computing* (accepted, in press). [\[arXiv\]](#)
- Eslik, D.*, **Kesgin, B.U.*** & Teğin, U. (2026). Recurrent neural networks implemented through spatiotemporal light propagation in optical fibers. Under review, *Optica*. [\[arXiv\]](#)
- Eslik, D., **Kesgin, B.U.** et al. (2026). Multimode fiber laser cavities as nonlinear optical processors. Under review, *Communications Physics*. [\[arXiv\]](#)
- Eslik, D., **Kesgin, B.U.** & Teğin, U. (2026). Low-cost passive single-shot ultrafast imaging at 685 Gfps. Under review, *Optics Letters*. [\[arXiv\]](#)
- Çarpınhoğlu, B., **Kesgin, B.U.** & Teğin, U. (2025). Real-time surrogate modeling of nonlinear pulse evolution in multimode fibers. Under review, *Optics Letters*. [\[arXiv\]](#)

* Equal contribution.

Patents

Kesgin, B.U. & Teğin, U. (2023). Method and system for enhanced machine learning utilizing chaotic attractors. International Patent (PCT) [WO2025063908A1](#).

Conference Presentations

1. **Kesgin, B.U.** & Teğın, U. (2025). Spatiotemporal chaos-based photonic neural networks. *SPIE Photonics West OPTO*, San Francisco, CA. [\[DOI\]](#)
2. Çarpınhoğlu, B., **Kesgin, B.U.** & Teğın, U. (2025). AI-assisted control over spatiotemporal nonlinear fiber dynamics. *SPIE Photonics West OPTO*, San Francisco, CA. [\[DOI\]](#)
3. **Kesgin, B.U.** & Teğın, U. (2025). Chaotic photonic reservoir computing with complex dynamics in multimode fibers. *CLEO 2025*, Long Beach, CA. [\[DOI\]](#)
4. **Kesgin, B.U.**, Yüce, F. & Teğın, U. (2025). Diffractive deep neural networks via mechanical mode coupling optimization. *CLEO 2025*, Long Beach, CA. [\[DOI\]](#)
5. **Kesgin, B.U.** & Teğın, U. (2025). Photonic neural networks with multimode fibers at the edge of spatiotemporal chaos. *CLEO Europe 2025*, Munich, Germany. [\[DOI\]](#)
6. **Kesgin, B.U.**, Yüce, F. & Teğın, U. (2025). Diffractive deep neural networks with multimode fibers through mode coupling optimization. *SPIE Optics+Photonics ETAI*, San Diego, CA. [\[DOI\]](#)
7. **Kesgin, B.U.** & Teğın, U. (2025). Chaotic photonic neural networks at the edge of spatiotemporal chaos. *SPIE Optics+Photonics ETAI*, San Diego, CA. [\[DOI\]](#)
8. Çarpınhoğlu, B., **Kesgin, B.U.** & Teğın, U. (2025). Convolutional neural networks for predicting nonlinear pulse propagation in multimode fibers. *Frontiers in Optics*, Denver, CO. [\[DOI\]](#)
9. **Kesgin, B.U.**, Yüce, F. & Teğın, U. (2026). Diffractive deep neural networks with multimode fibers. *SPIE Photonics West OPTO*, San Francisco, CA. [\[DOI\]](#)
10. Eşlik, D., **Kesgin, B.U.** & Teğın, U. (2026). Passive photonic recurrent neural networks via spatiotemporal multimode fiber dynamics. *SPIE Photonics West OPTO*, San Francisco, CA. [\[DOI\]](#)
11. Eşlik, D., **Kesgin, B.U.** & Teğın, U. (2026). Single-shot imaging achieving 685 billion FPS using spatial multiplexing. *SPIE Photonics West LASE*, San Francisco, CA. [\[DOI\]](#)

Honors, Awards & Fellowships

TÜBİTAK Intern Researcher Program (STAR) Fellow	2025
Koç University College of Engineering Dean's Honor Roll (×4)	2024–2026
Turkish Ministry of Youth and Sports National Athlete Scholarship	2022–2027

Teaching Experience

Teaching Assistant, EEE 429: Fundamentals of Optics	Spring 2025
Teaching Assistant, EEE 206: Electromagnetic Theory	Fall 2025

Technical Skills

Experimental: ultrafast fiber laser sources, free-space and fiber optical setups, SLM-based wavefront control, alignment and characterization.

Programming: Python (PyTorch, TensorFlow), C, C++, C#, MATLAB, R.

Tools: L^AT_EX, Git, Blender, Unity Engine; Arduino, Raspberry Pi.

Languages: Turkish (native), English (IELTS 7.5/9), German (basic), Spanish (basic).

Leadership and Athletic Experience

Turkish Volleyball Federation — National Beach Volleyball Athlete	Nov 2019 – Sept 2022
• Competed in 1 World, 3 European, and 4 Balkan Championships (35+ international matches); 17th globally at U19 World Championships.	
Koç University Debate Club — Board Member, Captain & Coach	July 2023 – June 2024
• Led tournament organization and member training; national tournament finalist.	